Physics-Grounded Urban Remote-Sensing Equations

Aerial and satellite sensing • Optical, thermal, LiDAR, SAR, InSAR, photogrammetry • Skill reference

Purpose. A compact, implementation-oriented reference for calibrating, correcting, simulating, and interpreting remote-sensing measurements in complex urban environments. Equations are provided in text notation to support direct transfer into analytical pipelines, model documentation, and SKILL.md libraries.

Core principle. Treat cities as three-dimensional, heterogeneous, directional scenes: roofs, façades, roads, vegetation, glazing, vehicles, shadows, and atmospheric effects frequently invalidate flat, Lambertian, or homogeneous-pixel assumptions.

1. Symbols and Constants

Symbol	Meaning	Typical SI unit
DN	Digital number / quantized detector output	dimensionless
L_lambda	Spectral radiance	W m^-2 sr^-1 um^-1
E_lambda	Spectral irradiance	W m^-2 um^-1
rho_lambda	Spectral reflectance	dimensionless
epsilon_lambda	Spectral emissivity	dimensionless
tau_lambda	Atmospheric/material transmittance	dimensionless
theta_s, theta_v	Solar and sensor-view zenith angles	radians or degrees
lambda	Optical wavelength	m or um
lambda_r	Radar wavelength	m
sigma_0, gamma_0	Radar backscatter coefficients	dimensionless or dB
c	Speed of light	2.99792458 x 10^8 m s^-1
h	Planck constant	6.62607015 x 10^-34 J s
k	Boltzmann constant	1.380649 x 10^-23 J K^-1
sigma	Stefan-Boltzmann constant	5.670374419 x 10^-8 W m^-2 K^-4

2. Radiometric Calibration

Digital number to at-sensor spectral radiance

```
L_lambda = M_L Q_cal + A_L
```

Use the calibration coefficients distributed with the image product. Q_cal is pixel DN; M_L and A_L are the radiance multiplicative and additive scaling factors. A gain/offset form is L_lambda = g_lambda DN + b_lambda. Apply coefficients per band, gain setting, acquisition mode, and processing level.

Photon response and signal-to-noise

```
N_e proportional to t_int integral[L_lambda R_lambda lambda/(hc) d_lambda]
SNR = N_e / sqrt(N_e + N_dark + N_read^2)
sigma_L_lambda approximately L_lambda / SNR
```

Low-radiance surfaces such as narrow streets, deep urban shadow, dark asphalt, and water have poorer SNR than bright concrete or sunlit roofs.

3. Passive Optical Physics

Surface energy conservation and Kirchhoff relation

```
alpha_lambda + rho_lambda + tau_lambda = 1
For opaque surfaces: tau_lambda approximately 0; therefore alpha_lambda + rho_lambda approximately 1
```

```
At local thermodynamic equilibrium: epsilon_lambda = alpha_lambda; hence epsilon_lambda
approximately 1 - rho_lambda
```

The opaque-surface approximation is often useful for roofs, asphalt, concrete, and soil. Glazing, water, thin materials, and wavelength-dependent material behavior require more careful treatment.

Lambertian reflection

```
L_lambda = rho_lambda E_lambda / pi
```

```
rho_lambda = pi L_lambda / E_lambda
```

This is an ideal diffuse-surface model. Glass, metal, solar panels, wet pavements, vehicle roofs, and directional building materials may show strong non-Lambertian behavior.

Solar irradiance on horizontal and inclined urban surfaces

```
E_dir,lambda = E_0,lambda d^-2 tau_sun,lambda cos(theta_s)
```

```
E_dir,lambda = E_0,lambda d^-2 tau_sun,lambda max(0, cos(i))
```

```
cos(i) = cos(theta_s)cos(beta) + sin(theta_s)sin(beta)cos(phi_s - phi_a)
```

Here beta is surface slope, phi_s is solar azimuth, and phi_a is outward surface-normal azimuth. For vertical façades set beta = 90 degrees. This is central to roof-energy mapping, façade illumination, and shadow-aware classification.

Top-of-atmosphere reflectance

```
rho_TOA,lambda = pi L_lambda d^2 / [E_0,lambda cos(theta_s)]
```

TOA reflectance normalizes radiance for solar geometry and Earth-Sun distance, but does not remove atmospheric scattering/absorption, adjacency effects, shadow, or urban BRDF effects.

Surface reflectance radiative transfer approximation

```
L_sensor,lambda = L_path,lambda + tau_up,lambda[(E_dir,lambda +
E_diff,lambda)rho_surface,lambda/pi] + L_adj,lambda
```

```
rho_surface,lambda approximately pi(L_sensor,lambda - L_path,lambda - L_adj,lambda) /
{tau_up,lambda(E_dir,lambda + E_diff,lambda)}
```

L_path is atmospheric path radiance; L_adj represents atmospheric and optical spill-in from nearby pixels. Bright roofs, concrete, water, and sharp shadow boundaries can make adjacency effects material in dense cities.

Beer-Lambert attenuation and scattering

```
I_lambda(R) = I_lambda,0 exp(-tau_lambda)
```

```
tau_lambda = integral[beta_ext,lambda(s) ds]; beta_ext,lambda = beta_abs,lambda +
beta_sca,lambda
```

```
Rayleigh scattering: beta_Rayleigh(lambda) proportional to lambda^-4
```

```
Aerosol Angstrom relation: tau_a(lambda) = beta lambda^-alpha
```

Short visible wavelengths are more affected by molecular scattering; atmospheric correction is consequently especially important for blue-band urban mapping.

Bidirectional reflectance distribution function

```
f_r(theta_i,phi_i,theta_r,phi_r,lambda) = dL_r(theta_r,phi_r,lambda) /
dE_i(theta_i,phi_i,lambda)
```

BRDF captures directional reflectance. It is important for façades, roof slopes, roadway directionality, glazing, and solar modules. Normalize using a physically based or fitted BRDF model when comparing acquisitions with different illumination/view geometry.

4. Urban Geometry, Shadow, and Radiation

Building shadow geometry

```
L_shadow = H / tan(alpha_s), where alpha_s = 90 degrees - theta_s
```

```
phi_shadow = (phi_s + 180 degrees) mod 360 degrees
```

At low solar elevation, tall-building shadows become disproportionately long and can dominate optical observations in dense urban cores.

Sky-view factor

```
SVF = (1/(2pi)) integral_over_visible_sky[cos(theta) dOmega]
```

```
Idealized canyon approximation: SVF approximately cos(arctan(2H/W))
```

Low SVF implies a more enclosed canyon, less radiative cooling to the sky, complex diffuse lighting, and stronger importance of multiple reflections. For operational urban work, calculate SVF from a DSM or 3-D mesh rather than relying on idealized canyon geometry.

Radiative view factor

```
F_i_to_j = (1/A_i) integral_Ai integral_Aj [cos(theta_i)cos(theta_j)/(pi r^2)] V_ij dA_j dA_i
```

V_ij is visibility: 1 for mutually visible points and 0 for occluded points. View factors support physically based thermal and shortwave simulations in urban canyons.

Diffuse radiosity / multiple reflections

```
J_i = E_i + rho_i sum_j[F_j_to_i J_j]
```

```
J = (I - R F^T)^-1 E
```

Multi-bounce radiosity can materially improve synthetic urban imagery and energy modeling where façade-road-facade interreflection is significant.

5. Thermal Infrared and Urban Heat

Planck law, Wien law, and Stefan-Boltzmann law

```
B_lambda(T) = [2hc^2/lambda^5] / [exp(hc/(lambda kT)) - 1]
```

```
lambda_max T = 2.897771955 x 10^-3 m K
```

```
M = epsilon sigma T^4
```

A surface near 300 K peaks near 9.66 um, within the thermal-infrared region commonly used for land-surface-temperature retrieval.

Thermal surface-leaving and at-sensor radiance

```
L_surface,lambda = epsilon_lambda B_lambda(T_s) + (1-epsilon_lambda)L_down,lambda
```

```
L_sensor,lambda = tau_up,lambda[epsilon_lambda B_lambda(T_s) + (1-epsilon_lambda)L_down,lambda] + L_up,lambda
```

```
B_lambda(T_s) = {[(L_sensor,lambda - L_up,lambda)/tau_up,lambda] - (1-epsilon_lambda)L_down,lambda}/epsilon_lambda
```

Thermal retrieval requires atmospheric transmittance, upwelling radiance, downwelling radiance, and emissivity. Brightness temperature alone is not land-surface temperature.

Brightness temperature

```
T_B = K_2 / ln(K_1/L_lambda + 1)
```

K_1 and K_2 are sensor-specific thermal calibration constants. Use the product documentation and ensure radiance units match those constants.

Urban surface-energy balance

```
R_n = (1-alpha)K_down + L_down - L_up
```

```
L_up = epsilon sigma T_s^4 + (1-epsilon)L_down
```

```
R_n = H + LE + G + Q_F + Delta_Q_S
```

```
H = rho_air c_p (T_s - T_a) / r_ah
```

```
LE = lambda_v ET
```

Urban analysis often requires anthropogenic heat Q_F and heat-storage change Delta_Q_S. These can be important in dense commercial, industrial, transport, and nighttime heat-island contexts.

6. Spectral Indices

Vegetation, built-up, and water indices

```
NDVI = (rho_NIR - rho_Red) / (rho_NIR + rho_Red)
EVI = G(rho_NIR - rho_Red) / [rho_NIR + C1 rho_Red - C2 rho_Blue + L]
Common EVI coefficients: G = 2.5, C1 = 6, C2 = 7.5, L = 1
NDBI = (rho_SWIR - rho_NIR) / (rho_SWIR + rho_NIR)
NDWI = (rho_Green - rho_NIR) / (rho_Green + rho_NIR)
MNDWI = (rho_Green - rho_SWIR) / (rho_Green + rho_SWIR)
```

Use atmospherically corrected surface reflectance. Indices are useful features, not direct physical measurements of imperviousness, vegetation fraction, or water extent, especially with urban mixed pixels and shadow.

Broadband albedo estimate

```
alpha_broadband approximately sum_i(w_i rho_i) + b
```

Weights must be sensor-, bandpass-, and product-specific. Do not transfer coefficients indiscriminately across satellites, UAV cameras, atmospheric-correction methods, or viewing geometry.

7. LiDAR and Photogrammetry

LiDAR range and uncertainty

```
R = c Delta_t / 2
sigma_R = (c/2) sigma_t
```

Total point-position uncertainty also includes GNSS, IMU, boresight, lever-arm, scan-angle, surface-slope, and registration contributions.

Simplified LiDAR equation

```
P_r(R) = P_t(c tau_p/2)(A_r/R^2) beta(R) T^2(R) eta
I_corr proportional to I_raw R^2 / [T^2(R) cos(theta_i)]
```

LiDAR intensity is not automatically reflectance. Correct interpretation depends on range, incidence angle, atmospheric loss, footprint geometry, receiver effects, gain, saturation, and calibration.

Ground sampling distance

```
GSD = H p / f
```

For an oblique aerial camera, effective GSD varies by image position, object elevation, viewing angle, and surface orientation.

Photogrammetric collinearity and stereo depth

```
x - x_0 = -f[r11(X-Xs)+r12(Y-Ys)+r13(Z-Zs)] / [r31(X-Xs)+r32(Y-Ys)+r33(Z-Zs)]
y - y_0 = -f[r21(X-Xs)+r22(Y-Ys)+r23(Z-Zs)] / [r31(X-Xs)+r32(Y-Ys)+r33(Z-Zs)]
Z = fB/d; sigma_Z approximately (Z^2/(fB)) sigma_d
nDSM = DSM - DTM
```

Depth uncertainty rises quadratically with range. An nDSM represents above-ground height and must be paired with semantic classification to separate buildings from trees and other elevated objects.

8. Synthetic Aperture Radar

Wavelength and monostatic radar equation

```
lambda_r = c/f
P_r = P_t G_t G_r lambda_r^2 sigma / [(4pi)^3 R^4 L]
```

The R^-4 dependence reflects spreading losses on both outbound and return paths. Radar response in cities is highly geometry dependent: building-wall/ground double bounce can be very bright, while radar shadow and layover can obscure targets.

Backscatter conventions

```
sigma_0 = sigma/A_ground
```

```
beta_0 = sigma/A_slant
gamma_0 = sigma_0/cos(theta_i)
sigma_0,dB = 10 log10(sigma_0)
```

Product definitions may vary by processor. Preserve metadata and harmonize terrain correction, incidence-angle treatment, and dB/linear conventions before comparing scenes.

SAR resolution and speckle

```
delta_r = c/(2B)
delta_az approximately L_a/2
ENL = mu_I^2/sigma_I^2
```

B is chirp bandwidth; L_a is antenna azimuth length; ENL is equivalent number of looks. Urban areas are rarely statistically homogeneous because of strong scatterers and complex geometric effects.

Polarimetric covariance

```
k = [S_HH, sqrt(2)S_HV, S_VV]^T
C = <k k^dagger>
```

Polarimetry can help characterize surface, volume, and double-bounce scattering mechanisms, subject to calibration and adequate spatial averaging.

9. InSAR and Deformation

Interferometric phase composition

```
phi_InSAR = phi_flat + phi_topo + phi_defo + phi_atm + phi_orbit + phi_noise
```

Line-of-sight displacement

```
d_LOS = (lambda_r/(4pi)) Delta_phi_defo
One 2pi phase cycle corresponds to Delta_d_LOS = lambda_r/2
```

InSAR retrieves line-of-sight motion, not automatically vertical subsidence or horizontal displacement. Separate viewing geometries and/or additional constraints are required for vector decomposition.

Height of ambiguity and coherence

```
h_amb = lambda_r R sin(theta_i)/(2B_perp)
gamma = |<s_1 s_2*>| / sqrt(<|s_1|^2><|s_2|^2>)
```

Persistent urban reflectors often retain coherence well, but construction, atmospheric variability, radar shadow, layover, temporal decorrelation, and changing moisture can limit performance.

10. Geometric and Terrain Corrections

Cosine correction and C-correction

```
rho_corr = rho_obs cos(theta_s)/cos(i)
rho_corr = rho_obs[cos(theta_s)+C]/[cos(i)+C], with C = b/m
```

Use with caution in cities. A simple illumination correction can become unstable or physically misleading for façades, cast shadows, occlusions, directional materials, and multiple reflections.

Orthorectification and relief displacement

```
x_map = Pi(R[X_ground - C])
d_r approximately r h/H
```

Use a DSM rather than only a terrain model for dense urban orthorectification. Tall buildings farther from nadir exhibit stronger apparent relief displacement.

11. Implementation Workflow

1. Ingest metadata: acquisition time, sensor response, calibration values, viewing geometry, solar angles, thermal constants, radar parameters, and processing level.

2. Convert raw measurements to physical quantities: radiance, reflectance, brightness temperature, LiDAR range/intensity, or calibrated SAR backscatter.

3. Model atmosphere: estimate transmittance, path radiance, aerosol/water-vapour effects, diffuse irradiance, and thermal upwelling/downwelling terms.

4. Use urban 3-D geometry: derive surface orientation, building/terrain occlusion, solar incidence, cast shadow, sky-view factor, and local radar incidence.

5. Retrieve target variables: reflectance, albedo, emissivity, LST, height, vegetation, imperviousness features, backscatter, coherence, or LOS deformation.

6. Quantify uncertainty: calibration, atmosphere, geometry, material emissivity/BRDF, co-registration, spatial mixing, and temporal mismatch.

7. Validate at matching scale/time using field radiometers, thermal loggers, weather stations, GNSS control, LiDAR checkpoints, building-height references, or deformation observations.

12. Urban-Specific Cautions

• **Mixed pixels:** a pixel can contain roofs, road, vegetation, façade, water, and shadow; retrieved values are radiance-weighted mixtures.

• **Cast shadow:** low reflectance may be an illumination state, not a dark material. Model visibility and direct/diffuse irradiance.

• **Glass and metal:** expect glint, directional reflectance, saturation, and potentially low or variable thermal emissivity.

• **Thermal anisotropy:** satellite LST is directional radiometric surface temperature, not pedestrian-level air temperature.

• **SAR geometry:** layover and radar shadow are common around high-rise structures; brightness is not simply material type.

• **Time sensitivity:** solar geometry, traffic, air-conditioning exhaust, moisture, construction, and cloud/aerosol conditions can change rapidly.

13. Model Selection Matrix

Objective	Recommended model family	Principal quantities
Surface reflectance	Atmospheric radiative transfer	TOA reflectance; Beer-Lambert; path/diffuse/adjacency radiance
Urban illumination & shadow	Solar geometry + ray tracing	Surface incidence; visibility; shadow length; SVF
Urban heat	Thermal transfer + energy balance	Planck; emissivity; atmospheric terms; Rn balance
Synthetic urban imagery	3-D RT / radiosity	BRDF; view factors; multi-bounce transfer
Building/tree height	LiDAR / stereo photogrammetry	Time-of-flight; collinearity; disparity; nDSM
SAR built-form mapping	SAR scattering + terrain geometry	Radar equation; sigma_0; gamma_0; polarimetry
Ground deformation	InSAR time series	Phase; coherence; LOS displacement; baseline

14. Source and Use Notes

This reference consolidates standard physical formulations used across optical radiative transfer, thermal remote sensing, LiDAR, photogrammetry, SAR, and InSAR. Apply the exact conventions and coefficients specified by each sensor provider and processor; equation forms, units, spectral response functions, and product definitions can differ materially.

Urban remote sensing benefits from explicit 3-D radiative-transfer and geometric modeling. High-fidelity workflows should integrate a DSM/mesh, material priors, acquisition geometry, atmospheric state, and uncertainty propagation rather than rely solely on per-pixel empirical transformations.

Evidence basis: ESA thermal remote-sensing instructional material; atmospheric and LiDAR equation references; urban 3-D radiative-transfer literature and DART urban/natural-environment simulation resources, as identified in the prior research step.